

⇒ Receiver Operating characteristics (ROC) and AUC

↓
Binary classification
↓
prob-score



Area under curve

X	Y	$\hat{y}$
x1	1	0.95
x2	1	0.92
x3	0	0.80
x4	1	0.76
x5	1	0.71

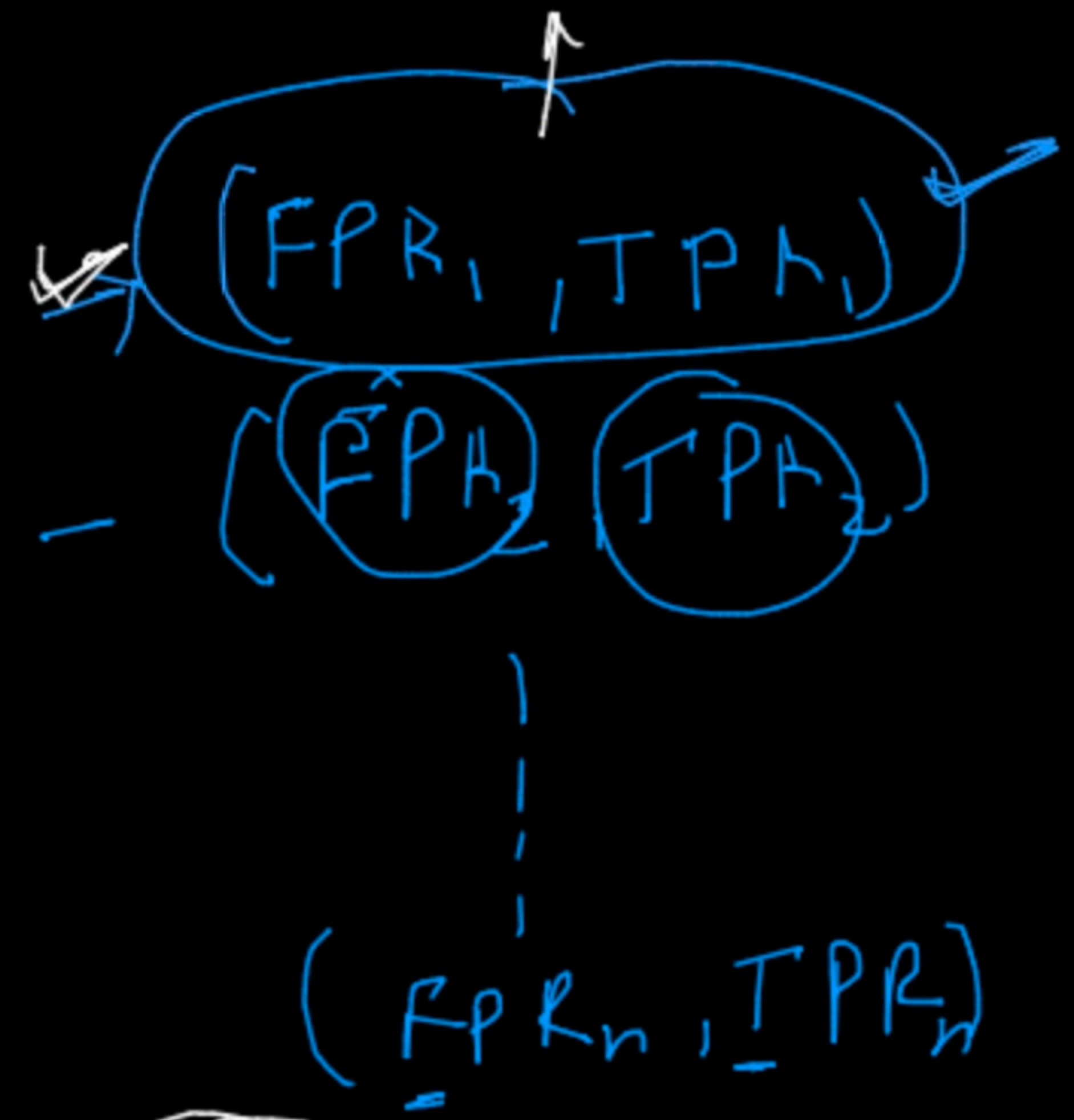
Step 1 → sort prob-score in descending order

skewed... → dataset

Actual Prediction

n -unique (0.4, 0.8)

X	Y	$\hat{Y}$	0.75	0.9	0.80
X1	1	0.45	1	1	1
X2	1	0.91	0	1	1
X3	0	0.50	0	0	1
X4	1	0.76	0	0	0
X5	1	0.71	0	0	0



Step 2 take threshold ≥ 0.95

(FPR, TPR)

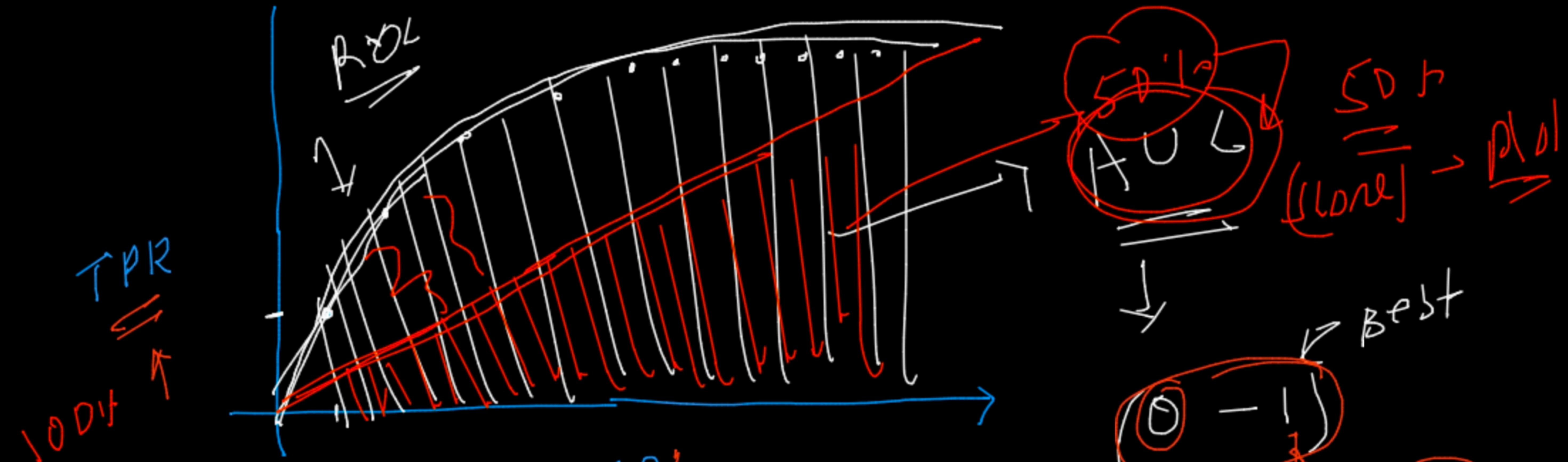
if prob-score ≥ 0.95 → 1

else → 0

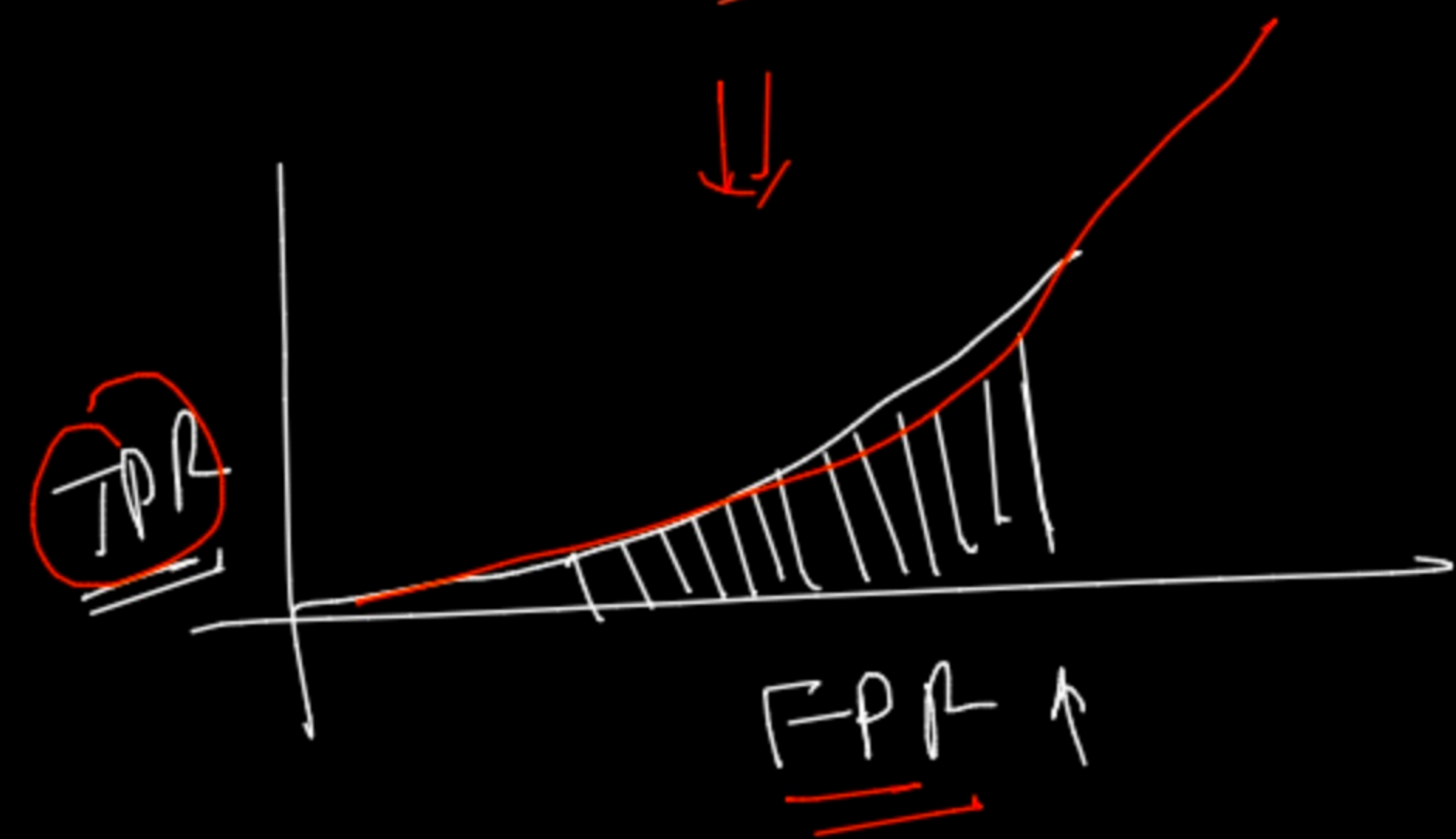
~~0.90~~

0.81

Step 3



Bad model



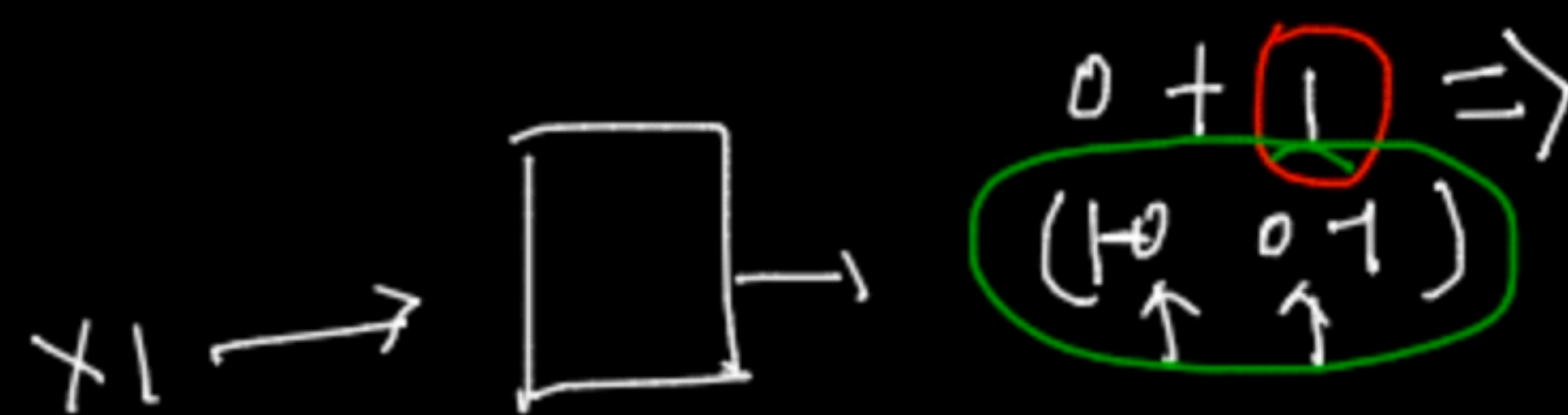
Best
worst

0.15

dumb model
OK

'Random model'

Log-loss → used your prob-score



$0.9 < 1$
 $\rightarrow 1 - 0.9 \rightarrow 0.1$

Binary classification

X	Y	$\hat{y} = P(Y=1)$	
(X1)	1	0.1	$0.9 \rightarrow -\log(0.9) * 1 + (1-1) * \dots$
(X2)	1	0.4	$0.6 \rightarrow -\log(0.6) * 1 \rightarrow 0.22 \uparrow$
X3	0	0.9	$0.1 \rightarrow -\log(0.1) * 0 - \log(0.9) * 1 \rightarrow 0.0457$
X4	0	0.6	$0.4 \rightarrow \dots$

formula

$$\log\text{-loss} = -\frac{1}{n} \sum_{i=1}^n \left\{ \log(p_i) * y_i + (1-y_i) * \log(1-p_i) \right\}$$

$\begin{matrix} & P \\ \left. \begin{matrix} \rightarrow \\ \end{matrix} \right\} & Q \end{matrix}$